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SVD Problem Solutions

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$$1. A^T A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$$

$$\begin{vmatrix} 20-\lambda & 10 \\ 10 & 5-\lambda \end{vmatrix} = \lambda^2 - 25\lambda = \lambda(\lambda-25) = 0 \rightarrow \lambda = 25, 0$$
$$\rightarrow \sigma = 5, 0$$

$$\lambda = 25: \begin{bmatrix} -5 & 10 \\ 10 & -20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = 2x_2 \rightarrow \vec{x} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{Normalize: } \vec{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda = 0: \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_2 = -2x_1 \rightarrow \vec{x} = x_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\text{Normalize: } \vec{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\vec{u}_1 = \frac{1}{\sigma_1} A \vec{v}_1 = \frac{1}{5} \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} = \frac{1}{5\sqrt{5}} \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{u}_2 = \text{unit basis for } N(A^T): \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = -2x_2$$

$$\rightarrow \vec{x} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} \rightarrow \vec{u}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\text{So } A = U \Sigma V^T = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix}$$

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$$2. B^T B = \begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix} \rightarrow \lambda = 1, 16, \sigma = 4, 1 \quad (2)$$

$$\lambda = 16: \begin{bmatrix} -15 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = 0 \rightarrow \vec{x} = x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \vec{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\lambda = 1: \begin{bmatrix} 0 & 0 \\ 0 & 15 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_2 = 0 \rightarrow \vec{x} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{u}_1 = \frac{1}{\sigma_1} A \vec{v}_1 = \frac{1}{4} \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{u}_2 = \frac{1}{\sigma_2} A \vec{v}_2 = \frac{1}{1} \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{So } A = U \Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$3. C^T C = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

$$\begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = \lambda^2 - 10\lambda + 16 = (\lambda-2)(\lambda-8) = 0 \rightarrow \lambda = 2, 8$$

$$\sigma = \sqrt{8}, \sqrt{2}$$

$$\lambda = 8: \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = x_2 \rightarrow \vec{x} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = 2: \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = -x_2 \rightarrow \vec{x} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\vec{u}_1 = \frac{1}{\sigma_1} A \vec{v}_1 = \frac{1}{\sqrt{8}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{u}_2 = \frac{1}{\sigma_2} A \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{So } A = U \Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$



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$$4. D^T D = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = \lambda^2 - 3\lambda + 1 = 0 \rightarrow \lambda = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

$$\sigma = \sqrt{\frac{3 \pm \sqrt{5}}{2}} = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

These are just the eigenvalues of D (one is the Golden Ratio)

$$= \phi, 1-\phi$$

$$\lambda = \phi^2: \begin{bmatrix} 2 - (\frac{3+\sqrt{5}}{2}) & 1 \\ 1 & 1 - (\frac{3+\sqrt{5}}{2}) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = \left(\frac{1+\sqrt{5}}{2}\right)x_2 \Rightarrow \vec{x} = x_2 \begin{bmatrix} \phi \\ 1 \end{bmatrix}$$

$$\lambda = (1-\phi)^2: \begin{bmatrix} 2 - (\frac{3-\sqrt{5}}{2}) & 1 \\ 1 & 1 - (\frac{3-\sqrt{5}}{2}) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = \left(\frac{1-\sqrt{5}}{2}\right)x_2$$

$$\rightarrow \vec{x} = x_2 \begin{bmatrix} 1-\phi \\ 1 \end{bmatrix} \rightarrow \vec{v}_2 = \frac{1}{\sqrt{(1-\phi)^2 + 1}} \begin{bmatrix} 1-\phi \\ 1 \end{bmatrix}$$

$$\left(\frac{1-\sqrt{5}}{2}\right)^2 + 1 = \frac{5-\sqrt{5}}{2}$$

$$\vec{u}_1 = \frac{1}{\phi} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{\phi^2+1}} \begin{bmatrix} \phi \\ 1 \end{bmatrix} = \frac{1}{\phi\sqrt{\phi^2+1}} \begin{bmatrix} 1+\phi \\ \phi \end{bmatrix} = \frac{1}{\sqrt{\phi^2+1}} \begin{bmatrix} \phi \\ 1 \end{bmatrix} = \vec{v}_1$$

$$\vec{u}_2 = \frac{1}{1-\phi} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{(1-\phi)^2+1}} \begin{bmatrix} 1-\phi \\ 1 \end{bmatrix} = \frac{1}{(1-\phi)\sqrt{(1-\phi)^2+1}} \begin{bmatrix} 2-\phi \\ 1-\phi \end{bmatrix} = \vec{v}_2$$

SVD decomposition is $D = Q \Lambda Q^T$ (i.e., $U = V = Q$)

because D is already symmetric.